Can Agents Trust a Replication Verifier? Epistemic Contracts for Scientific Agent Tools

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***Abstract*—AI agents that do science increasingly hand the question "can this finding be trusted?" to a verifier tool. Such tools are judged on accuracy, not on the guarantees agents depend on. We propose a testable methodology for *epistemic contracts*: explicit guarantees about what a tool's output may reflect (temporal validity, isolation from the queried paper's labels), which decisions it supports, and how it is audited. We apply it to SCIRELIGRAPH, an MCP replication verifier whose contract tests reach a verified mutation score of 61% (83% after tests for surviving mutants). On 274 SCORE claims with replication outcomes, a verifier meeting the integrity requirements discriminates only modestly (AUROC 0.627), and citation graphs bring no established improvement. LLM judges rank competitively but cannot be certified as temporally valid, and a decoy probe caught two of four naming outcomes for never-replicated papers. In an agent-in-the-loop experiment with placebo and tool-unavailable controls, a bare probability pushed most agents into near-chance automatic decisions they avoided when told the tool was unavailable. The contract removed most of these errors for three of four agents, beyond a generic warning; the fourth misread it. In a pre-specified follow-up, wordings that state the permitted action fixed this for all four agents, and with a synthetic decision-ready verifier agents automated only the claims the contract licensed (at most 4% of the others, vs. 64–78% with a bare number). Agents follow stated actions, so a licence is a policy that must be explicit and tested against its readers.**

***Index Terms*—agentic AI, epistemic contracts, AI for science, verification, LLM-as-a-judge, contract testing, metamorphic testing, data leakage, MCP, replicability**

I. INTRODUCTION

Systems that automate parts of research [1] keep having to decide which earlier findings are solid enough to build on. They can hand that decision to a *verifier* tool, for instance an LLM acting as a judge [2]. The stakes are real. Large replication projects found that many published findings in the social and behavioural sciences do not hold up [3], [4], [5], and the DARPA SCORE program has released outcomes for 274 replicated claims from 164 papers, of which 151 (55.1%) replicated [6], [7]. There are also reasons for caution about delegating. LLMs that predict experimental outcomes cannot tell which of their predictions are reliable [8], and AI-led teams reproduced far fewer published results than human-only or AI-assisted teams [9].

When an agent asks a verifier about claim c at time t, it quietly assumes three things: that the answer uses only information available at t, that the probability means what it says, and that the answer can be audited later. We call these guarantees an *epistemic contract*. Security and correctness checks ask whether a tool is safe and behaves as specified; an epistemic contract states what the tool's *output may reflect* and which decisions it *licenses*. None of the ingredients is new on its own. Design by contract [10], model documentation [11], calibration [12], metamorphic [13] and mutation testing [14], and tool-use evaluation [15] all have long histories. Our contribution is a *testable methodology* that combines them for agent tools, together with evidence of a failure mode that only shows up when they are combined. These guarantees also fail silently. A verifier fitted on bibliometric data scraped today can absorb signals from after the replication, an LLM judge may simply remember the outcome [16], and random splits can put claims from the same paper in both training and test data [17], [18].

We ask two related questions: *if a verifier is built and tested against an explicit epistemic contract, is it good enough for an agent to rely on, and does the agent use it correctly?* We expected time-bounded citation graphs (H1), corrected for non-independent support (H2), to improve forecasts, and lifting the evidence cutoff to inflate apparent accuracy (H3). The data supported none of the three, and we report these negative results in full. The agent experiment, by contrast, changed our view of what a contract must cover. One agent read the tool's honest warning that its score was weak as evidence *against the claim* and rejected 96% of the claims it saw (§VII). A contract therefore has to be tested against its *consumers*, not only against the tool.

Contributions.

1) *A testable epistemic-contract methodology for verifier tools* (§III): five requirements, a typing of features by when they become available, and a reusable JSON Schema with tool-agnostic test relations. Typing and its date checks caught two leaks in our own pipeline, one of which would otherwise have produced our best-looking result.
2) *Contract tests with verified adequacy* (§IV–§V): an MCP tool that enforces the contract on every call, metamorphic and isolation tests, and verified mutation analysis.
3) *A decoy-calibrated memorization probe for LLM judges* (§VI-F), which separates recall from fabrication; two of four models named outcomes for never-replicated papers.
4) *Evidence of a gap between contract and consumer* (§VII). A bare verifier probability induced errors that most agents avoided when the tool was reported un-

available; the contract removed most of them for three agents but misled the fourth. A pre-specified follow-up shows that stating the permitted action reliably steers all four agents, including towards automation when a tool is good enough to act on.

II. Background and Related Work

Replication prediction. Prediction markets [19] classified 103 pooled replication outcomes with 73% accuracy [20], and for SCORE, human forecasters reached a median AUC of 0.76 on 37 high-powered replications [21]. Machine-learning forecasters built on statistical and textual features also report above-chance accuracy [22], [23], [24]. SCORE's own machine assessments [25] were mostly built to predict *human* judgements and did not discriminate replication outcomes [26]. Other work adds algorithmic traders to markets [27], benchmarks agents that *carry out* replications [28], or screens reproducibility with LLMs [29]. We differ in three ways: we use only realised outcomes as labels, we enforce temporal validity, and we evaluate the verifier as a tool that an agent acts on. Nonreplicable findings are known to be cited more [30]; we ask whether citation *structure* helps where raw counts do not. Unlike fact-checking [31], replication forecasting predicts a future study, which exposes it to hindsight leakage.

LLM judgements and agent tools. Evaluations of LLM forecasters are vulnerable to temporal leakage [32]. GLEAN turns trajectory evidence into calibrated correctness probabilities for high-stakes agent verification [33], and claim-level calibration makes LLM confidences usable for decisions, with post-hoc temperature and Platt scaling as baselines [34]. Work on MCP evaluates multi-step tool use [15], explores tool spaces with metamorphic checks [35], defends the tool layer at run time [36], and records agents' actions during experiment reproduction [37]. These studies ask whether agents *use* tools correctly or safely. Other work shows how agents mishandle what tools return: they adopt corrupted returns at high rates, and reliability labels barely change this [38]. Uncertainty gets lost at handoffs unless it is carried explicitly [39], and passing through a tool boundary does not by itself give a statement epistemic warrant [40]. Agents also struggle to know when not to act [41] and how much time has passed [42]. Existing tool contracts constrain calls and actions, through preconditions and effects [43] or behavioural invariants [44]. Ours constrain what an *output* may reflect and license, and we test whether agents read it correctly (Table I). In SE for ML, hidden data dependencies [45] and data management [46] are known problems and datasheets make provenance explicit [47]; MCP [48] exposes a predictor as a tool but says nothing about what its output may be trusted for.

TABLE I

Positioning against the closest work, by each work's stated scope. ✓: a stated focus; ○: addressed in a different sense; –: not a stated focus (not a claim of absence). Our agent evaluation is single-turn, hence ○ in (e). Columns: (a) realised outcomes as labels; (b) temporal leakage controlled; (c) calibration or selective accuracy evaluated; (d) the tool's output contract tested; (e) agent use of the output evaluated.

Work	(a)	(b)	(c)	(d)	(e)
Replication predictors [22], [23], [24]	✓	–	–	–	–
SciPredict [8]	✓	✓	✓	–	–
GLEAN [33], CLAIM-CAL [34]	✓	–	✓	–	○
ReplicatorBench [28]	✓	–	–	–	✓
MCP agent benchmarks [15], [35], [37]	–	–	–	○	✓
MCP runtime defence [36]	–	–	–	○	–
Tool contracts [43], [44]	–	–	–	○	○
Tool overreliance [38]	–	–	–	–	✓
This work	✓	✓	✓	✓	○

III. A Verifier Contract for Agentic Pipelines

A. Problem definition

Let c be an original empirical claim, t a forecast cutoff, and $\mathcal{E}(c,t)$ the admissible evidence dated $\leq t$. Let $Y_c \in \{0,1\}$ be the outcome of a later independent replication under a predeclared criterion. The verifier returns $\hat{p}_c = \hat{P}(Y_c = 1 \mid c, \mathcal{E}(c,t))$ with an evidence record. We evaluate discrimination (AUROC), probabilistic accuracy with strictly proper scores (Brier, log loss) [49], calibration [12], and selective accuracy [50]. Selective accuracy matters because an agent may act only on its confident outputs.

B. Requirements of the epistemic contract

R1 Temporal validity. No feature or evidence of the queried claim may encode information dated after t, in particular replication results, replication-paper metadata, or post-cutoff citations. (The model is necessarily trained on *other* claims' later outcomes, so `as_of` filters evidence, not training labels; a `labels_as_of` field, added after the agent experiments, states this. Our evaluation is therefore retrospective: it excludes post-cutoff evidence about the queried claim and within-paper label leakage, but does not show that an equivalent verifier could have been trained at the cutoff.) Calls whose t falls after the outcome may be public are refused. **R2 Label validity.** The label is the declared criterion, reconciled to published counts; alternative criteria are reported as sensitivity analyses. **R3 Honest generalisation.** Claims from the same paper are never split across training and test data, and a claim is never scored by a model that saw any label from its own paper. **R4 Decision-relevant output.** The output must beat the base rate on proper scores and must expose the base rate. If it is used for gating, its confident outputs must be more accurate than its unconfident ones. **R5 Auditability.** Every prediction must be traceable to a source archive (with checksum), an evidence access date, the feature tiers used, and a model configuration. R1 and R3 constrain what the output may *reflect*, R4 what it *licenses*, and R2 and R5 make both checkable. R1, R2, R3 and R5 are integrity requirements that can be enforced. R4 is different. It is an empirical acceptance criterion, and a verifier may simply fail it. When that happens, the response must *expose* the failure through its base rate, skill and gating verdict, rather than suggest that the predictor is ready for decisions. We therefore distinguish three levels: a verifier is *integrity-compliant* if it meets R1–R3 and R5, *decision-qualified* if it also meets R4, and *consumer-validated*

TABLE II
REQUIREMENTS, ENFORCEMENT MECHANISMS, AND WHAT ENFORCEMENT CAUGHT OR SHOWED IN THIS STUDY.

Req.	Enforcement mechanism	Finding in this study
R1 temporal	Feature tiers; dated-evidence filter; assertion that the cutoff precedes the earliest replication start; call-time refusal; metamorphic test	Excluded undated TOP Factor scores; exposed the achieved-size leak (D^*) and post-launch design registration (D)
R2 label	Build-time assertion of 274/164/151; alternative criteria kept as separate labels	Exact reconciliation; three sensitivity labels
R3 generalisation	Paper-grouped CV; nested grouped tuning; leave-paper-out in the tool; label-flip test	Unregularised starter baseline scored worse than the base rate
R4 decision use	Proper scores vs. base rate; reliability diagram (artifact); selective accuracy; `selective_ok` flag	B1 Brier skill 2.6% (CI includes 0); no harness verifier shows a supported gating gain
R5 audit	Manifest (sha256, access date, tiers, missingness); per-call audit log; scripted tables and numbers	All reported numbers regenerate from outputs

if its output has been tested against the agents that consume it (R6, §VIII). In our results, R1–R3 and R5 are met by construction and tested (§V). R4's first clause is met only nominally (B1 Brier skill 2.6%, paired CI includes zero), and its gating clause is not met by any harness verifier (§VI-E).

C. Feature-availability typing

R1 is enforced by *typing* every feature with the moment it becomes knowable. **Tier A** (original publication): sample size; p-value and whether it is reported exactly; marginal significance ($0.01 < p \leq 0.05$); converted effect size $|r|$; power to detect half the original effect; interaction vs. main effect; test-statistic and analysis family; publication year; and discipline. **Tier T** (original publication): title, abstract and keywords. **Tier G(t)** (cutoff-dependent): the evidence graph of §IV. **Tier D** (replication registration): new vs. secondary data, the preregistered power target, and the *planned* sample size implied by that target. **Tier D^*** (after data collection): the *achieved* replication sample size and the power computed from it. A forecast "at publication" may use A, T and G(t); once the replication design is registered, a forecast may add D. SCORE registered designs after the project started (99.6% of claims, median 94 days later), so Tier D precedes the outcome but not the launch. D^* is never admissible and appears only as a leakage illustration: the achieved size exceeds the planned size by a median factor of 1.33 and can depend on how data collection went. Typing also forced us to decide about fields whose timing the archive does not record. SCORE's journal-level TOP Factor scores look like publication metadata, but their rating date is not stored. Since R1 cannot be verified for them, we exclude them.

IV. THE SCIRELIGRAPH HARNESS

Table II maps each requirement to its enforcement point: the label gate (R2), the evidence cutoff (R1) and the tool boundary (R1, R3–R5). The implementation is 2877 lines of Python and runs end to end from public sources.

Data and E0 gate. The harness downloads the publication-frozen SCORE analyst package [7] and records its SHA-256. It selects version-of-record, non-COVID replications and *asserts* that they reproduce the published totals of 274 claims, 164 papers and 151 successes [6]. Of the plausible selections we tried (for example, 427 reports, or 309 claims including COVID-era studies), only this one reproduces the published totals. The set spans 54 journals, publication years 2009–2018 and six fields; 128 replications used new data and 146 secondary data.

Evidence and cutoffs. For each original paper, resolved by DOI (164/164), the harness caches the OpenAlex [51] record and every citing work: 23,666 in total, with authorships, venues, topics and references. Time filtering is applied only at feature construction. The *strict* cutoff, 2018-12-31, is asserted to precede the earliest replication start in the archive (2019-08-06); 8,268 citing works fall before it. The *launch* cutoff is the day before each replication project started; it is known for 259 claims, and the remainder use the strict cutoff. The *none* cutoff admits everything available at access time (October 2026) and is a diagnostic only.

Evidence graph G(t). For claim c, the graph contains the original paper and all citing works dated $\leq t$, with *cites*, *shares-author* (dependency) and *shares-reference* (coupling) edges. From it we derive three feature groups. *Flat evidence* covers citation counts and rates and the reference count. *Graph structure* covers citing-venue entropy, field diversity, the share of cross-field citers, citer–citer density and coupling. *Dependency correction* covers the self-citation share, the number of independent support groups (connected components of non-self citers under shared authorship) and the number of distinct citing authors. The graph is purely bibliographic, with no semantic supports/contradicts edges, and its features are defined per paper.

Models and protocol. B0 predicts the training-fold prevalence. Tabular models use median imputation with missingness indicators, standardisation and one-hot encoding (rare levels pooled), followed by L2 logistic regression. The regularisation strength $C \in \{0.003, \ldots, 1\}$ is chosen by inner grouped 3-fold CV on log loss. A gradient-boosted tree baseline (depth 2, 150 iterations) has fixed hyperparameters. The text model is TF-IDF (1–2-grams) with tuned logistic regression. We use stratified, paper-grouped 5-fold CV repeated over 20 seeds, and report the mean and SD over repetitions. For inference we average each claim's out-of-fold predictions over seeds and draw 2,000 paper-clustered bootstrap resamples. Paired differences are computed on identical resamples, and their p-values are Holm-adjusted over the family of 11 planned comparisons; reported CIs are unadjusted, so a CI can exclude zero while p_{Holm} is large. The **permutation null** permutes all graph and dependency features across papers within publication-year strata, 30 times, re-running the full CV each time. This preserves marginal distributions and the year–citation relation while breaking the link to outcomes.

V. THE VERIFIER AS AN AGENT TOOL

Offline evaluation tells us how good a verifier is, but not whether the deployed tool behaves the way it was

```
// verify_claim("Ej3y_single-trace", as_of="2018-12-31")
{"claim_id": "Ej3y_single-trace",
  "stage": "publication", "as_of": "2018-12-31",
  "labels_as_of": "2025-12-15", "p_replicate": 0.4673,
  "base_rate": 0.5531, "cv_auroc": 0.62,
  "cv_brier_skill": 0.0277, "selective_ok": false,
  "n_evidence": 8, "tiers_used": ["A", "G"],
  "n_train_claims": 273,
  "drivers": [{"feature": "g_cross_field_frac",
  "logit_contribution": 0.119}],
  "provenance": {"archive_sha256": "80a221f0deb7...",
  "model_config": "81f3fdc8b3c61867"},
  "advice": "weak signal: do not gate actions
  on this score"}
// same claim, as_of="2023-01-01"
{"error": "contract_violation",
  "detail": "as_of 2023-01-01 is on/after replication
  launch 2021-05-24: evidence may reflect the outcome"}
```

Listing 1. Real `verify_claim` responses over MCP (hashes truncated).

evaluated. We therefore implemented the verifier as a library with an MCP [48] server exposing a single tool, `verify_claim(claim_id, as_of, stage)`, and enforced the contract inside the call path. (1) *R1:* for every call, the graph features of *all* claims are recomputed from the raw cache with evidence filtered to `as_of`, and calls on or after the claim's replication launch are refused. (2) *R3:* the model is refitted without any claim from the queried paper (leave-paper-out), with the same nested tuning as the harness. (3) *Stage gating:* `publication` uses tiers A+G, and `launch` uses A+D but answers only once the design is registered. On SCORE, registration never precedes the project start, which the tool already treats as too late, so it refuses every launch-stage call. (4) *R4:* the response carries the base rate, the cross-validated AUROC and Brier skill of that configuration, and a `selective_ok` flag that is true only if the lower 95% bound of top-decile accuracy exceeds overall accuracy in the selective-accuracy test (§VI-E); it is false for every harness configuration we evaluated. With about 27 claims in the top decile, the rule passes only if that decile is at least about 81% accurate, so it is deliberately conservative. (5) *R5:* each call appends a record with inputs, outputs, archive hash and model-configuration hash to an audit log. Listing 1 shows the actual responses an MCP client received in our smoke test for the publication-stage configuration (A+G), whose skill differs from B1's: one valid call and one refused call.

Contract tests and their adequacy. Because the "right" probability is unknown, we test *relations* that must hold under the contract (Table III). A *metamorphic* test injects citing works dated after `as_of` into every paper's evidence and requires a bit-identical output, while a positive control checks that earlier works *are* counted. An *isolation* test flips the labels of every claim from the queried paper and requires an unchanged prediction. A *gating* test randomises Tier-D values and requires an unchanged publication-stage output. Further tests cover refusals, malformed calls, the base rate and audit records. A JSON Schema that any verifier can adopt fixes the shape of the response, and schema-conformance tests check answers and refusals against it. To see whether the tests would catch real violations, we built 5 mutants, each removing one guarantee: evidence filtering, leave-paper-out, stage gating, the launch refusal or audit logging. The unmodified tool passes all 9 original tests, and every mutant is killed (5/5).

Independent mutation analysis. Since we wrote these mutants ourselves, we also ran a standard tool, `mutmut` 2.5.1, over the verifier module (251 mutants; Table IV). Its own verdicts missed some kills, so we re-ran every reported survivor in isolation with a freshly restored source file, and 19 turned out to be killed. The remaining survivors pointed to real gaps, which we closed in two rounds. The first added ten tests for the exact output schema, the true branch of the gating rule, evidence and training counts, and audit hashing. The second added four: two for the temporal boundary that R1 depends on (a work dated exactly t is admitted, one dated $t{+}1$ is not) and two oracles (the launch-date fallback, and loaded evidence matching the raw cache). Because these fourteen tests were written after we saw the survivors, the final score is optimistic, and we report it next to the pre-analysis 61%. None of the 42 remaining survivors threatens a contract guarantee: 13 change hyperparameters that the contract leaves open, 9 touch the command-line entry point, 8 alter the untested "drivers" output, 5 change wording or rounding, and 7 are equivalent or nearly so.

VI. Results on the SCORE Testbed

Replication forecasting is a good testbed for the contract. It comes with realised outcomes, a natural temporal boundary, and a decision that an agent would plausibly delegate. We first ask how good an integrity-compliant verifier can be (A–E), then how LLM judges behave under the same contract (F).

A. How good is a careful verifier?

Table V summarises the main configurations. The metadata model B1, which uses only information from the original study, reaches AUROC 0.627±0.017 (CI [0.55, 0.71]). (B0 scores below 0.5 only because grouped CV shifts the training prevalence away from each test fold's own.) The evidence of signal is fragile, however. When we refit the whole protocol inside 200 paper-bootstrap resamples, so that fitting and tuning variability is included, B1's interval widens to [0.46, 0.72], with 93% of replicates above 0.5. This interval is conservative, since each resample holds only about 63% distinct papers. The probabilistic gain is small: a Brier score of 0.2409 against 0.2474, a skill of 2.6%, with a paired CI that includes zero. A gradient-boosted variant ranks claims about as well (0.620) but has a worse Brier score (0.2483), and the TF-IDF text model is at chance (0.481). Finally, an unregularised starter baseline did *worse* than the base rate (Brier 0.267 vs. 0.247), a reminder that base-rate comparison belongs in a verifier's acceptance tests. Under three alternative success criteria, B1 stays between 0.60 and 0.64, and within fields it is close to chance in psychology and health (0.53, n=94).

TABLE III

CONTRACT TESTS FOR THE VERIFIER TOOL AND MUTATION ANALYSIS. EACH MUTANT REMOVES ONE GUARANTEE FROM THE IMPLEMENTATION; × MARKS THE TESTS THAT FAIL (KILL THE MUTANT). THE UNMODIFIED VERIFIER PASSES ALL 9 TESTS.

Req.	Contract test	leaky	no-LPO	no-gate	no-refuse	no-audit
R1	Metamorphic: citing works dated after `as_of` added to every paper leave output unchanged	×				
R1	Positive control: works dated before `as_of` are counted					
R1	Call on/after the replication launch is refused				×	
–	Malformed date or unknown stage is refused (2 cases)					
R3	Flipping all labels of the queried claim's paper leaves its prediction unchanged		×			
R1	Publication stage: Tier-D values have no influence			×		
R4	Base rate equals training prevalence; weak-signal advice when selective test fails		×			
R5	Each call appends an audit record with provenance hashes					×
Mutant killed?		yes	yes	yes	yes	yes

TABLE IV

MUTATION SCORE OF THE VERIFIER MODULE (`mutmut`, 251 MUTANTS), BY TEST SUITE. EACH VERIFIED ROW RE-RUNS ALL SURVIVING MUTANTS IN ISOLATION. LATER TESTS (SCHEMA CONFORMANCE, DESIGN REGISTRATION) AND THE `LABELS_AS_OF` FIELD POSTDATE THESE RUNS.

Stage	Tests	Killed	Score
Tool's own report	9	134	53%
Verified: original contract tests	9	153	61%
+ round 1 (schema, gating rule, counts, audit)	19	204	81%
+ round 2 (cutoff boundary, oracles)	23	209	83%
excluding 7 equivalent mutants	23	209	86%

B. H1/H2: Do evidence graphs and dependency correction help?

Not detectably. Flat citation evidence added to metadata changes AUROC by -0.013 (unadjusted CI $[-0.027, -0.001]$; $p_{\text{Holm}} = 0.34$ after adjustment for 11 comparisons). Adding graph structure (M1 vs. B4: $+0.013$, CI $[-0.028, +0.046]$) or dependency correction (M2 vs. M1: $+0.000$, CI $[-0.017, +0.017]$) changes nothing, and M2 is indistinguishable from metadata alone ($+0.000$, CI $[-0.041, +0.033]$). The permutation null tells the same story. Over 30 within-year permutations, the real graph block reaches 0.632 against a null mean of 0.605 ($p = 0.10$): perhaps a weak signal, but not enough to pay for its variance (B1 alone: 0.639 on the same seeds). Graph features alone reach 0.591, and raw pre-cutoff citation counts are uninformative (0.52), in line with [30].

C. H3: Does removing the cutoff inflate performance?

Not detectably for these features. Admitting all evidence available in 2026 raises the mean $\log(1+\text{citations})$ from 3.30 to 4.50 but does not improve discrimination ($M2_\infty$ vs. M2: -0.010, CI $[-0.059, +0.035]$; G_∞ vs. G: -0.030, CI $[-0.107, +0.050]$). With so little signal, this test has low power. It does not justify dropping the cutoff, and it says nothing about LLM verifiers [16].

D. Design-stage information and two leaks we caught

Moving the cutoff to each replication's start date leaves performance unchanged ($M2_\ell$: 0.620). Our first design-stage model used the archive's replication sample size and power, and it looked like our best model. Typing showed that these are *achieved* values, known only after data collection. With the planned design alone (Tier D), the gain disappears (D vs. B1: -0.003, CI $[-0.037, +0.034]$); adding the achieved values back (D^*) recovers part of it ($+0.024$, CI $[-0.005, +0.053]$), not significantly but in the direction post-hoc information would. Checking dates revealed a second, subtler leak: the planned design is fixed at registration, a median 94 days *after* the project started, so our original "launch-time" label for D was wrong. We relabelled it and made the tool refuse launch-stage calls before registration. A field's *name* did not reveal when it became available; typing and dates did.

E. Can an agent gate on confidence?

A natural agent policy is to act only when the verifier is confident. When we rank claims by $|p - 0.5|$, B1's most confident 10% are correct only 0.44 of the time (CI [0.26, 0.68]), compared with 0.63 overall (D: 0.48 vs. 0.64). M2's top decile does better (0.67 vs. 0.59), but its interval [0.38, 0.82] is far too wide to support gating, and mean predictions per quintile span a narrow range (reliability diagram in the artifact). No harness configuration meets R4's gating clause; §VII shows what this means under a review budget.

F. LLM-as-judge verifiers

An agent could also put the question to an LLM, so we evaluated four LLMs from two vendors under the same protocol. Each vendor contributed one model whose vendor-stated training data end before the SCORE results paper (April 2026 [6]) and one after it: `gpt-5.4-mini-2026-03-17` and `claude-haiku-4-5-20251001` (2025), `gpt-6.1-sol` and `claude-opus-5-5` (April and June 2026). A system prompt dated at the strict cutoff asked for the probability that a high-powered replication would succeed (one sample per claim, lowest reasoning setting). There were three conditions: (i) *identified*: title, citation, abstract and the Tier-A statistics; (ii) *de-identified*: field and statistics only, which removes the cues needed to recognise the paper; (iii) a *memorization probe* asking directly whether the paper was replicated in SCORE and with what outcome, also asked about 100 *decoys*: journal-matched papers that SCORE sampled but never replicated or reproduced. All responses parsed, and the judging calls cost $5.68.

TABLE V

OUT-OF-FOLD FORECASTING PERFORMANCE ON 274 SCORE CLAIMS (164 PAPERS). AUROC: MEAN±SD OVER 20 REPETITIONS OF PAPER-GROUPED 5-FOLD CV, WITH PAPER-CLUSTERED BOOTSTRAP 95% CI OF THE SEED-AVERAGED PREDICTIONS. BRIER: MEAN OVER REPETITIONS (LOWER IS BETTER). BSS: BRIER SKILL SCORE VS. B0. CUTOFF "REGIST.": USES THE DESIGN REGISTERED AFTER THE REPLICATION STARTED; "POST": INFORMATION AVAILABLE ONLY AFTER DATA COLLECTION (LEAKAGE ILLUSTRATION). SELECTED CONFIGURATIONS; ALL OTHERS (GBT, TEXT-ONLY, GRAPH-ONLY, PERMUTED, LAUNCH-CUTOFF M2) ARE IN THE ARTIFACT.

ID	Features	Cutoff	AUROC	95% CI	Brier	BSS
B0	Training-fold prevalence	strict	0.492±0.001	[0.37, 0.56]	0.2474	+0.0%
B1	Original-study metadata, LR	strict	0.627±0.017	[0.55, 0.71]	0.2409	+2.6%
B4	B1 + flat citation counts	strict	0.613±0.019	[0.54, 0.69]	0.2432	+1.7%
M1	B4 + graph structure	strict	0.624±0.018	[0.55, 0.70]	0.2406	+2.8%
M2	M1 + dependency correction	strict	0.620±0.020	[0.55, 0.70]	0.2406	+2.8%
D	B1 + planned replication design	regist.	0.625±0.019	[0.54, 0.71]	0.2429	+1.8%
D*	D + achieved size/power	post	0.638±0.024	[0.58, 0.72]	0.2403	+2.9%
$M2_\infty$	M2	none	0.614±0.017	[0.53, 0.71]	0.2409	+2.7%

TABLE VI

LLM-AS-JUDGE VERIFIERS ON THE SAME CLAIMS. *Identified*: TITLE, CITATION, ABSTRACT AND ORIGINAL STATISTICS; *de-identified*: FIELD AND ORIGINAL STATISTICS ONLY. AUROC WITH PAPER-CLUSTERED 95% CI; Δ VS. B1 ON IDENTICAL CLAIMS AND RESAMPLES. PROBE: SHARE ANSWERING THAT THE PAPER WAS REPLICATED IN SCORE (*aware*) FOR REAL SCORE PAPERS VS. JOURNAL-MATCHED DECOYS THAT WERE NEVER REPLICATED, AND ACCURACY OF NAMED OUTCOMES (COUNTED WHENEVER AN OUTCOME IS NAMED, REGARDLESS OF STATED AWARENESS) VS. THE MAJORITY-OUTCOME BASELINE ON THE SAME CLAIMS. BRIER (PLATT): AFTER PLATT SCALING FITTED ON TRAINING FOLDS ONLY (PAPER-GROUPED CV, 20 REPETITIONS). B1 HERE IS SCORED ON ITS SEED-AVERAGED PREDICTIONS, SO ITS BRIER DIFFERS SLIGHTLY FROM THE PER-SEED MEAN IN TABLE V.

Model	Input	AUROC	95% CI	ΔAUROC vs. B1	95% CI	Brier	Brier (Platt)	Top-10% acc.
Claude Haiku 4.5 (2025-07)	identified	0.566	[0.48, 0.65]	-0.061	[-0.154, +0.026]	0.2541	0.2466	0.52
	de-identified	0.592	[0.51, 0.68]	-0.036	[-0.132, +0.057]	0.2561	0.2479	0.70
Claude Opus 5.5 (2026-06)	identified	0.655	[0.57, 0.74]	+0.027	[-0.044, +0.103]	0.2392	0.2323	0.89
	de-identified	0.620	[0.53, 0.70]	-0.007	[-0.097, +0.077]	0.2478	0.2393	0.70
GPT-5.4 mini (2025-08)	identified	0.600	[0.52, 0.68]	-0.028	[-0.106, +0.050]	0.2753	0.2438	0.70
	de-identified	0.570	[0.48, 0.66]	-0.058	[-0.160, +0.038]	0.2758	0.2473	0.56
GPT-6.1 sol (2026-04)	identified	0.663	[0.58, 0.74]	+0.036	[-0.031, +0.105]	0.2594	0.2313	0.78
	de-identified	0.612	[0.53, 0.70]	-0.016	[-0.107, +0.073]	0.2627	0.2413	0.74
B1 (metadata LR)	same claims	0.628	[0.55, 0.71]	–	–	0.2387	–	0.44

Memorization probe

Model	Aware (real)	Aware (decoy)	Named outcomes	Named acc.	Majority
Claude Haiku 4.5 (2025-07)	0.0%	0.0%	0/274	–	–
Claude Opus 5.5 (2026-06)	0.0%	0.0%	3/274	0.67	0.67
GPT-5.4 mini (2025-08)	42.7%	24.0%	77/274	0.64	0.56
GPT-6.1 sol (2026-04)	2.2%	2.0%	6/274	1.00	0.83

Discrimination and probabilistic accuracy (Table VI). The two post-cutoff models had the highest AUROCs of any verifier in this study (GPT-6.1 sol 0.663, CI [0.58, 0.74]; Claude Opus 5.5 0.655, CI [0.57, 0.74]), without a statistically established advantage over B1 (+0.036 and +0.027). The pre-cutoff models were below B1 (0.600 and 0.566). The quality of the raw probabilities varies sharply. Opus 5.5's raw Brier score is as low as B1's (0.2392 vs. 0.2387; Brier mixes ranking and calibration, so this does not establish equal calibration), whereas the other three are worse than the base rate (0.2594, 0.2753, 0.2541; B0 0.2474). We applied the same in-fold Platt scaling, a standard post-hoc calibrator [12], [34], to every judge. This brings GPT-6.1 sol to 0.2313 and Opus 5.5 to 0.2323 (Brier skill +6.5% and +6.1%), the lowest of any verifier, while leaving AUROC almost unchanged. Neither recalibrated judge is distinguishable from B1 (ΔBrier CI [−0.0229, +0.0074] and [−0.0258, +0.0132]). The pre-cutoff judges improve only marginally (0.2466 and 0.2438 vs. B0 0.2474). Since a monotone recalibration improves their Brier scores without changing their ranking, the post-cutoff judges mainly fail on calibration, though recalibration needs labelled outcomes, so the judge is no longer zero-shot.

Memorization and fabrication. Removing identifying information lowers the AUROC point estimate for both post-cutoff models (GPT-6.1 sol +0.051, CI [+0.008, +0.101]; Opus 5.5 +0.035, CI [−0.024, +0.092]; outside the Holm family). This is suggestive rather than conclusive. GPT-5.4 mini shows a similar drop (+0.030), and de-identification also removes the abstract, which is legitimate information. Both post-cutoff models rarely recall outcomes explicitly: they named outcomes for only 6 and 3 of 274 claims (GPT-6.1 sol's were all correct). Their gain from identification may therefore come from implicit memorization or from a legitimate reading of the abstract, and the two cannot be told

TABLE VII

AGENT-IN-THE-LOOP TRIAGE ON IDENTICAL CLAIMS (SCENARIO A: 274 VALID QUERIES; B: 259 POST-LAUNCH QUERIES, WHERE THE UNPROTECTED TOOL SILENTLY RETURNS A NO-CUTOFF PREDICTION AND THE CONTRACT-AWARE TOOL REFUSES). U: AGENT SEES ONLY `p_replicate`; C: FULL CONTRACT OUTPUT. WRONG AUTO = ACCEPTED NON-REPLICATING PLUS REJECTED REPLICATING CLAIMS, AS A SHARE OF ALL CLAIMS; Δ = C$-$U WITH PAPER-CLUSTERED 95% CI. REVISED WORDINGS: TABLE IX.

	Agent	Esc. U	Esc. C	Wrong U	Wrong C	Δ wrong (pp)
A	GPT-5.4 mini	66%	93%	14.6%	3.6%	-10.9 [-16.4, -6.3]
	GPT-6.1 sol	81%	100%	9.1%	0.0%	-9.1 [-13.5, -5.1]
	Claude Haiku 4.5	59%	1%	17.9%	54.0%	+36.1 [+27.3, +44.4]
	Claude Opus 5.5	94%	100%	2.9%	0.0%	-2.9 [-5.2, -1.1]
B	GPT-5.4 mini	54%	100%	20.1%	0.0%	-20.1 [-29.0, -12.5]
	GPT-6.1 sol	75%	100%	12.0%	0.0%	-12.0 [-19.8, -5.6]
	Claude Haiku 4.5	49%	85%	21.6%	6.9%	-14.7 [-22.3, -7.5]
	Claude Opus 5.5	91%	100%	3.9%	0.0%	-3.9 [-7.5, -0.8]

apart from outside the model. R1 cannot be certified for these judges [32]. The probe also exposes fabrication. GPT-5.4 mini claims to know of a replication for 42.7% of real papers and 24.0% of decoys, and it names outcomes for 12 decoys that were never replicated. GPT-6.1 sol does so for 2, and the Claude models for none. Unlike instance-level contamination tests [52], decoys calibrate claimed recall against papers where any claimed outcome is false.

Confidence. Opus 5.5 is the only verifier whose top-decile gain has a CI excluding zero: 0.89 (CI [0.67, 1.00]) vs. 0.62, gain CI [+0.07, +0.38]. The result is nominal, since it is unadjusted across at least 11 selective comparisons, and it was obtained on identified input, where memorization would act (de-identified: 0.70). For GPT-6.1 sol the gain CI ([−0.00, +0.36]) touches zero. The best candidate for gating is thus a judge whose temporal validity cannot be certified.

VII. AGENT IN THE LOOP

Contract tests show that the tool *reports* its limits, not that an agent *uses* them [37]. To find out, we ran a minimal single-turn triage workflow. An LLM agent receives a claim and the output of `verify_claim`, and must *accept* the claim (the pipeline builds on it), *reject* it, or *escalate* it to a human reviewer with limited capacity. The system prompt is identical across workflows; only the tool output differs. The *unprotected* agent sees `p_replicate` alone; the *contract-aware* agent sees the full response. In scenario A, a valid query at the strict cutoff, both agents receive the leave-paper-out prediction for each of the 274 claims. In scenario B, the query is dated 2023-06-01, after the replication launched. Here the unprotected tool silently returns the no-cutoff prediction, while the contract-aware tool refuses; this covers 259 claims. Scenario B deliberately measures what an agent does when a contract violation is *silent* rather than *surfaced*, not the harm of leakage itself (§VI-C). All four LLMs served as agents (4,264 decisions, plus 5,191 in the control, robustness and revised-wording runs below; $11.10 in total).

Table VII shows the results. No verifier probability was extreme (range 0.30–0.76), so unprotected agents already escalated many claims, but the decisions they did make were close to chance, mostly rejecting claims that in fact replicated.

For three of the four agents, the contract turned these errors into escalations. Wrong automatic decisions changed by −10.9, −9.1 and −2.9 percentage points in scenario A (GPT-5.4 mini, GPT-6.1 sol and Opus 5.5), and the same agents escalated every post-launch refusal in scenario B.

TABLE VIII

CONTROL ARMS FOR SCENARIO A (274 CLAIMS): SHARE ESCALATED / SHARE OF WRONG AUTOMATIC DECISIONS. *Unavailable*: THE AGENT SEES THE CLAIM AND A NOTE THAT THE TOOL OUTPUT IS UNAVAILABLE (A CUE IN ITSELF). *Placebo*: `p_replicate` PLUS A GENERIC "THIS ESTIMATE IS UNCERTAIN" NOTE. *Contract*: FULL `verify_claim` OUTPUT. *Always escalate* NEEDS NO MODEL.

Agent	Unavailable	Placebo	Unprotected	Contract
GPT-5.4 mini	100% / 0.0%	82% / 8.8%	66% / 14.6%	93% / 3.6%
GPT-6.1 sol	100% / 0.0%	95% / 2.9%	81% / 9.1%	100% / 0.0%
Claude Haiku 4.5	98% / 0.4%	63% / 15.7%	59% / 17.9%	1% / 54.0%
Claude Opus 5.5	86% / 4.0%	96% / 1.8%	94% / 2.9%	100% / 0.0%
Always escalate	100% / 0.0%			

Control arms (Table VIII, scenario A). Told that the tool output is unavailable, agents mostly escalate (GPT-5.4 mini 100%, GPT-6.1 sol 100%, Haiku 4.5 98%). It is the bare probability that tempts them into automatic decisions: compared with the unavailable arm, wrong decisions rose by +14.6, +9.1 and +17.5 points (CIs [+9.8, +19.8], [+5.2, +13.5], [+11.9, +23.4]). A generic "this estimate is uncertain" note undoes part of the harm. For three agents the contract does more than that note, changing wrong decisions relative to the placebo by −5.1, −2.9 and −1.8 points (CIs [−9.2, −1.7], [−5.2, −1.1], [−3.6, −0.4]; unadjusted). This suggests that the contract's specific content matters, although the placebo is also much shorter (68 vs. 417 characters), so length and salience are not matched. The unavailable arm is itself a cue to escalate, and the contract beats it only for Opus 5.5 (−4.0, CI [−8.2, −0.8]), which accepted some claims in that arm; an always-escalate policy does just as well. For a verifier this close to chance, the contract's value is to take away the false authority of a bare number, not to improve automation, and fewer errors are not the same as better decisions: if one escalation costs c relative to one wrong automatic decision, the contract-aware agent beats the unprotected one only when c is below 0.42, 0.47 and 0.47 for these three agents, and never for Haiku 4.5.

Prompt robustness. Under an independently worded prompt (three agents), wrong decisions still order bare number > placebo > contract for both GPT agents (GPT-5.4 mini: 17.5%, 9.1%, 1.5%), Haiku 4.5's misuse persists (50.4% wrong), and magnitudes shift (GPT-6.1 sol's bare-number errors: 9% to 1.8%).

The fourth agent, Claude Haiku 4.5, *misused* the contract. Given the full response in scenario A, it rejected 96% of claims, and its wrong automatic decisions rose from 18% to 54%. Its stated reasons reveal two misreadings. It took the verifier's weak skill ("AUROC 0.62") and the advice "do not gate actions on this score" as evidence *against the claim*

TABLE IX

PRE-SPECIFIED R6 VALIDATION (SCENARIO A, 274 CLAIMS; WRONG = WRONG AUTOMATIC DECISIONS). *R6*: WORDINGS V2–V4 UNDER TWO SYSTEM PROMPTS (OPUS: ONE), WORST CELL SHOWN. *Decision-ready*: A SYNTHETIC VERIFIER (AUROC 0.88) WHOSE V3 OUTPUT LICENSES ACCEPT/REJECT FOR CONFIDENT SCORES; *out*: SHARE OF UNLICENSED CLAIMS DECIDED AUTOMATICALLY; *agree*: DECISIONS CONSISTENT WITH THE LICENCE. UNDER THE R6 CONTRACT ALL AGENTS ERR ON 4.7% OF CLAIMS, THE ERROR RATE OF THE LICENSED CLAIMS, FIXED BY THE TOOL.

	Orig.	R6	Decision-ready tool		
Agent	wrong	esc./wrong	bare: out	R6: out	R6: agree
GPT-5.4 mini	3.6%	100%/0.0%	78%	4%	98%
GPT-6.1 sol	0.0%	100%/0.0%	67%	0%	100%
Haiku 4.5	54.0%	100%/0.0%	76%	4%	98%
Opus 5.5	0.0%	100%/0.0%	64%	0%	100%

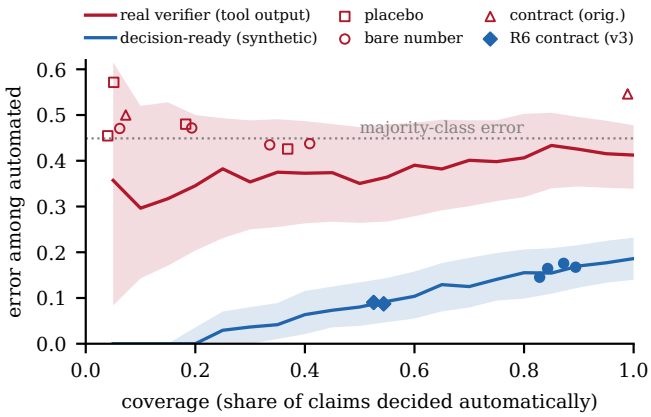


Fig. 1. Risk–coverage. Lines: threshold policy on each tool's scores (automate the k claims with largest $|p - 0.5|$, decide by $p \geq 0.5$; 95% bands). Points: observed agent decisions at self-chosen coverage, no budget imposed (hollow: real tool; filled: decision-ready).

rather than against the tool, and it read `selective_ok` as a warning about "selective reporting". It also rejected 15% of the refusals in scenario B, as if a refusal were evidence against the claim. An exploratory revision (v2), designed *after* this failure, renamed the field `confidence_gating_supported`, made the advice action-oriented and said that a refusal carries no information about the claim; Haiku 4.5 then escalated 100% of claims in both scenarios. A correct tool, in other words, is not enough. Contract fields and advice are an interface in their own right, and they must be tested against the agents that read them.

Pre-specified validation of R6 (Table IX). For a fairer test, before making any call we froze two new wordings, the experimental conditions, a set of comprehension questions and a success criterion, and committed them to the public repository. Wording v3 separates statements about the claim from statements about the tool and lists the permitted actions. Wording v4 keeps the original fields, including `selective_ok`, and adds a single interpretation sentence. All four agents saw v2–v4 under the main prompt, and the three agents from the robustness run also saw them under prompt 2 (11,098 calls, $11.42). In all 21 agent–wording–prompt cells, the agents escalated every claim and made 0.0% wrong automatic decisions, which meets the pre-specified criterion. Haiku 4.5's misuse disappeared under both new wordings (v3 vs. original: −54.0 points, CI [−61.1, −46.3]), and the three agents tested on refusals escalated all of them. Asked directly about the *original* output, every agent answered that the score cannot justify accepting or rejecting the claim, that the licensed action is escalation, and that a refusal carries no information about the claim (all answers correct). So Haiku 4.5 understood the very output it misused; its only comprehension error was reading `selective_ok` as selective reporting (23% of answers; v4: 0%). The failure lay between understanding and acting. This is the knowing–doing gap described for LLM agents [53], [54], and it resembles agents that notice a conflict yet still adopt a corrupted tool return [38]. Stating the permitted action closes it, which shows that agents follow such statements, not that they understand the contract better.

Not just abstention. These results alone could mean the contract simply teaches agents to abstain. To test this, we gave the same agents a synthetic *decision-ready* verifier, calibrated by construction (AUROC 0.88). Its scores outside $[0.15, 0.85]$ cover 53% of claims and are 91% accurate, and its v3 output permits accepting or rejecting those claims and asks for escalation on the rest. With the contract, every agent automated all of the licensed claims and at most 4% of the others, agreeing with the licence on 98–100% of claims. Their identical 4.7% wrong decisions are the error rate of the licensed claims themselves, fixed by the tool rather than by the agents. With the bare number, agents also automated 64–78% of the unlicensed claims and made 12.0–15.3% wrong decisions. Asked whether the score alone justifies a decision, agents told licensed and unlicensed claims apart (92% correct, scored per claim for all agents: 'yes' for the 12 licensed claims of 30, 'no' otherwise). The frozen key wrongly expected 'yes' for all 30 and gives 32%; no behavioural result uses either key, and both are in the artifact. The contract therefore steers agents towards automation where its licence permits it, not only towards abstention; the licence thresholds themselves are a policy choice encoded in the tool. We stress that this is a controlled interface experiment. It tests whether agents follow a licence, not whether a real verifier can earn one, and it says nothing about how well replications can be forecast.

Decision value under a review budget. When review capacity is limited, some claims have to be decided automatically, and the sensible use of a score is to automate the most confident ones. Fig. 1 evaluates that threshold policy on the scores themselves; it is not an agent run under an enforced budget. For the real verifier, the error among automated decisions stays between 30% and 45% at every coverage. At escalation budgets of 25, 50 and 75%, it decides 29.9%, 17.5% and 9.6% of all claims wrongly, against 10.6%, 4.0% and 0.7% for the decision-ready tool. Agents given the bare real score sit at or above that curve. No wording can move the curve, because the contract adds no information about individual claims; under a budget, its value is to say up front that the review budget, not

the tool, must absorb the risk.

VIII. Discussion: Implications for Agentic Systems

Machine verifiers are weak signals, and apparently weaker than human crowds. Our best leakage-safe verifiers reach an AUROC of about 0.63 with a Brier skill below 3%, while human forecasters reached a median AUC of 0.76 on SCORE [21] (on different evaluation sets, so the comparison is only indicative). An orchestrator should not prune hypotheses or literature on such a score, but it can use it to decide which claims to check first, shown with its base rate and evidence record. LLM judges, whose cutoff cannot be enforced on their weights, should be wrapped in the same contract, with decoy probes and recalibration.

Make temporal validity a type, and test it at the tool boundary. Typing caught two leaks, and metamorphic tests give regression-tested assurance that the deployed tool honours the evaluated cutoff. Tool schemas should carry `as_of`, `labels_as_of`, the base rate, the gating verdict and provenance.

Add a consumer-side clause to the contract. The same output moved three agents towards escalation and misled the fourth. We therefore propose a sixth requirement, **R6 Consumability**: the output must state the action it permits (for example, "escalate") under an explicit policy, not only the actions it rules out, and the contract must be validated against the agents that will read it. R6 is a *semantic* counterpart of consumer-driven contract testing [55], which tests consumers' expectations of an interface; here we test LLM consumers' *interpretation* of the output. Reliability labels barely changed reliance on corrupted tool returns in a different task [38]; in ours, where escalation was an explicit option, reliability fields already helped three agents, and R6 wordings steered all four. This shows that agents *follow* stated actions, not that they could derive them from error rates. Whether to automate depends on the cost of escalation relative to error (§VII), which the tool cannot know, so a contract should separate what the tool guarantees (error rates by score band) from the policy that turns them into actions, and declare that policy. A licence also carries authority and is thus an injection channel [36]; agents should treat it as advice bounded by their caller's policy.

Ship negative controls, contract tests and audits with the tool. The permutation null, mutation analysis and a filesystem audit (which found two paper identifiers differing only in case sharing a cache entry) all caught problems and belong in a verifier's continuous-integration suite.

IX. Threats to Validity and Limitations

Data and construct. With 274 claims in 164 clusters, our nulls rule out large gains, not small ones. A failed replication does not prove a claim false, and SCORE's selection of papers, which our cutoff does not control, may not carry over to other fields. OpenAlex data are imperfect, and our graph has no semantic edges, so H1 and H2 are tested only on bibliographic links. **Models.** We used simple, auditable models. The LLM judges used one prompt and the agents two, with one sample per claim and each model's lowest reasoning setting. Cutoffs are vendor-stated, and the pre/post-cutoff split is confounded because individual SCORE reports appeared before the consolidated paper. The probe measures only what a model will *say* it knows. Refitting inside bootstrap resamples widens B1's AUROC interval to [0.46, 0.72] (§VI-A). **Tool and agent.** The agent experiment is single-turn and offline, with default decoding and one sample per decision, so run-to-run variability is not in the CIs. Controls cover scenario A only, the placebo is not length-matched, and no agent ran under a binding review budget. Accepting a non-replicating claim and rejecting a replicating one count equally as errors; most errors were rejections. v3 and v4 were pre-specified, but the author wrote all the wordings. Because the R6 wordings say "escalate", their 100% escalation partly reflects instruction following; the decision-ready control shows that agents also follow a licence to act, but that tool is synthetic. The methodology is shown on one tool class only; a second tool is future work. H1–H3 were not preregistered, and apart from the pre-specified R6 follow-up, analyses outside the Holm family are exploratory. A mutation score measures fault detection, not proof that the contract always holds.

X. Conclusion

A verifier tool for scientific agents needs an epistemic contract, tests that enforce it at the tool boundary, and evidence that agents read it as intended; each layer caught something the others missed. On SCORE, a verifier meeting the integrity requirements discriminates only modestly and cannot gate agent actions. In the loop, a bare probability led most agents into errors they avoided when the tool was reported unavailable. A contract that states the permitted action removed those errors for all four agents and, when the tool was good enough, licensed automation that the agents followed; such a licence is a policy, and should be declared as one. Replication forecasts should inform scientific agents, not gate them.

Artifact Availability

The reproducibility package (code, MCP server, tests, predictions, cached LLM and agent responses, and the scripts behind every table and number) is publicly available at https://github.com/luxinlabs/scireligraph (MIT licence). SCORE data are available from OSF [7].

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References

[1] C. Lu *et al.*, "The AI scientist: Towards fully automated open-ended scientific discovery," arXiv:2408.06292, 2024.

[2] L. Zheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, vol. 36, 2023, pp. 46 595–46 623.

[3] Open Science Collaboration, "Estimating the reproducibility of psychological science," *Science*, vol. 349, no. 6251, p. aac4716, 2015.

[4] C. F. Camerer *et al.*, "Evaluating replicability of laboratory experiments in economics," *Science*, vol. 351, no. 6280, pp. 1433–1436, 2016.

[5] ——, "Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015," *Nature Human Behaviour*, vol. 2, pp. 637–644, 2018.

[6] A. H. Tyner *et al.*, "Investigating the replicability of the social and behavioural sciences," *Nature*, vol. 652, no. 8108, pp. 143–150, 2026.

[7] Center for Open Science, "Materials for "Investigating the replicability of the social and behavioral sciences"," OSF project g5sny, analysis data and code package (15 Dec 2025), https://osf.io/g5sny/, 2025.

[8] U. M. Sehwag *et al.*, "SciPredict: Can LLMs predict the outcomes of scientific experiments in natural sciences?" in *Proc. ICML*, 2026, arXiv:2604.10718.

[9] A. Brodeur *et al.*, "AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science," *Proceedings of the National Academy of Sciences*, vol. 123, no. 22, p. e2524747123, 2026.

[10] B. Meyer, "Applying "design by contract"," *Computer*, vol. 25, no. 10, pp. 40–51, 1992.

[11] M. Mitchell *et al.*, "Model cards for model reporting," in *Proc. Conference on Fairness, Accountability, and Transparency (FAT*)*, 2019, pp. 220–229.

[12] C. Guo *et al.*, "On calibration of modern neural networks," in *Proc. ICML*, 2017, pp. 1321–1330.

[13] T. Y. Chen *et al.*, "Metamorphic testing: A review of challenges and opportunities," *ACM Computing Surveys*, vol. 51, no. 1, pp. 4:1–4:27, 2018.

[14] Y. Jia and M. Harman, "An analysis and survey of the development of mutation testing," *IEEE Transactions on Software Engineering*, vol. 37, no. 5, pp. 649–678, 2011.

[15] Z. Wu *et al.*, "MCPMark: A benchmark for stress-testing realistic and comprehensive MCP use," in *Proc. ICLR*, 2026.

[16] O. Sainz *et al.*, "NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark," in *Findings of EMNLP*, 2023, pp. 10 776–10 787.

[17] S. Kaufman *et al.*, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 15:1–15:21, 2012.

[18] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

[19] A. Dreber *et al.*, "Using prediction markets to estimate the reproducibility of scientific research," *Proceedings of the National Academy of Sciences*, vol. 112, no. 50, pp. 15 343–15 347, 2015.

[20] M. Gordon *et al.*, "Predicting replicability—analysis of survey and prediction market data from large-scale forecasting projects," *PLoS ONE*, vol. 16, no. 4, p. e0248780, 2021.

[21] F. Mody *et al.*, "Large-scale human predictions of the replicability of published social and behavioural science papers – a multi-study analysis," MetaArXiv preprint, https://osf.io/preprints/metaarxiv/vgyed_v1, 2026.

[22] A. Altmejd *et al.*, "Predicting the replicability of social science lab experiments," *PLoS ONE*, vol. 14, no. 12, p. e0225826, 2019.

[23] Y. Yang, W. Youyou, and B. Uzzi, "Estimating the deep replicability of scientific findings using human and artificial intelligence," *Proceedings of the National Academy of Sciences*, vol. 117, no. 20, pp. 10 762–10 768, 2020.

[24] W. Youyou, Y. Yang, and B. Uzzi, "A discipline-wide investigation of the replicability of psychology papers over the past two decades," *Proceedings of the National Academy of Sciences*, vol. 120, no. 6, p. e2208863120, 2023.

[25] N. Alipourfard *et al.*, "Systematizing confidence in open research and evidence (SCORE)," SocArXiv preprint, https://doi.org/10.31235/osf.io/46mnb, 2021.

[26] S. Rajtmajer *et al.*, "Innovations in machine assessment of replicability," MetaArXiv preprint, https://osf.io/preprints/metaarxiv/bdmte_v1, 2026.

[27] T. Chakravorti *et al.*, "Human-AI collaboration for estimating scientific replicability," in *Proc. International Conference on Hybrid Human-Artificial Intelligence (HHAI)*, 2026, arXiv:2605.27394.

[28] B. Nguyen *et al.*, "ReplicatorBench: Benchmarking LLM agents for replicability in social and behavioral sciences," in *Proc. ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026, pp. 11 693–11 704.

[29] T. Holtdirk *et al.*, "Automated reproducibility assessments in the social and behavioral sciences using large language models," arXiv:2606.13670, 2026.

[30] M. Serra-Garcia and U. Gneezy, "Nonreplicable publications are cited more than replicable ones," *Science Advances*, vol. 7, no. 21, p. eabd1705, 2021.

[31] D. Wadden *et al.*, "Fact or fiction: Verifying scientific claims," in *Proc. EMNLP*, 2020, pp. 7534–7550.

[32] D. Paleka *et al.*, "Pitfalls in evaluating language model forecasters," in *Proc. ICLR*, 2026.

[33] Y. Zhang *et al.*, "GLEAN: Guideline-grounded evidence accumulation for high-stakes agent verification," in *Proc. ICML*, 2026, arXiv:2603.02798.

[34] T. Abbasli *et al.*, "Claim-level confidence calibration for reliable decision making with large language models," in *Proc. 5th Workshop on Uncertainty Reasoning and Quantification in Decision Making (at ACM SIGKDD)*, 2026, arXiv:2608.22483.

[35] Z. Liu and Y. Zhang, "SGVEF-LOOP: Coverage-guided progressive topological exploration and fact-grounded metamorphic evaluation for MCP agents," in *Proc. ACL (Volume 1: Long Papers)*, 2026, pp. 26 573–26 599.

[36] S. Yergattikar, "Securing the tool layer: A threat taxonomy and runtime defense framework for model context protocol deployments," in *Proc. ACL (Volume 6: Industry Track)*, 2026, pp. 865–871.

[37] H. Hong *et al.*, "Overview of the NLPCC 2026 shared task 11: Agent-based experiment reproduction from scientific papers," arXiv:2609.11117, 2026.

[38] H. Yang *et al.*, "Agents' overreliance on unreliable tools," arXiv:2609.05587v2, 2026.

[39] K. Shi *et al.*, "Confidence laundering in agent systems: Why uncertainty needs a latent carrier," arXiv:2606.20662, 2026.

[40] O. Romanchuk and R. Bondar, "Semantic laundering in AI agent architectures: Why tool boundaries do not confer epistemic warrant," arXiv:2601.08333, 2026.

[41] X. Liu *et al.*, "AgentAbstain: Do LLM agents know when not to act?" arXiv:2607.10059, 2026.

[42] Y. Cheng *et al.*, "Your LLM agents are temporally blind: The misalignment between tool use decisions and human time perception," in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 37 082–37 104.

[43] R. Suresh Babu and L. Ganesh Iyer, "Contract2Tool: Learning preconditions and effects for reliable tool-augmented LLM agents," arXiv:2606.07904, 2026.

[44] V. P. Bhardwaj, "Agent behavioral contracts: Formal specification and runtime enforcement for reliable autonomous AI agents," arXiv:2602.22302, 2026.

[45] D. Sculley *et al.*, "Hidden technical debt in machine learning systems," in *Advances in Neural Information Processing Systems*, 2015, pp. 2503–2511.

[46] S. Amershi *et al.*, "Software engineering for machine learning: A case study," in *Proc. ICSE-SEIP*, 2019, pp. 291–300.

[47] T. Gebru *et al.*, "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.

[48] Anthropic, "Model context protocol specification," https://modelcontextprotocol.io, 2024.

[49] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.

[50] R. El-Yaniv and Y. Wiener, "On the foundations of noise-free selective classification," *Journal of Machine Learning Research*, vol. 11, pp. 1605–1641, 2010.

[51] J. Priem, H. Piwowar, and R. Orr, "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts," arXiv:2205.01833, 2022.

[52] S. Golchin and M. Surdeanu, "Time travel in LLMs: Tracing data contamination in large language models," in *Proc. ICLR*, 2024.

[53] T. Schmied *et al.*, "LLMs are greedy agents: Effects of RL fine-tuning on decision-making abilities," arXiv:2504.16078, 2025.

[54] S. Deb and A. Krishnan, "STOCKTAKE: Measuring the gap between perception and action in LLM agents with a fair oracle," arXiv:2607.13618, 2026.

[55] I. Robinson, "Consumer-driven contracts: A service evolution pattern," https://martinfowler.com/articles/consumerDrivenContracts.html, 2006.